MIL-C-48867B (AR)
31 March 1983
SUPERSEDING
MIL-C-48867A (AR)
5 October 1978

MILITARY SPECIFICATION

CHARGE, PROPELLING, M204
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements and quality assurance provisions for the loading, assembling and packing for the Charge, Propelling, M204 used on the Lightweight Company Mortar System (LWCMS).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, forms a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

FSC 1310

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

9312698	- Charge, Propelling, M204
9313703	- Packing and Marking for Box, Fiberboard, Outer for Charge, Propelling M204, or Container Assembly
9313727	- Packaging for Pallet for Box, Fiberboard, Outer for Charge, Propelling, M204 or Container Assembly

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Assembly. The propelling charge assembly shall comply with all requirements specified on drawing (dwg.) 9312698 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Gross weight of propelling charge. The weight of the loaded propelling charge shall not be greater than the assessed propellant weight plus 3.08 grams and not less than the assessed propellant weight plus 2.09 grams when tested as specified in 4.5.3.

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3.4 Velocity. The propelling charges shall impart a corrected muzzle velocity of not more than 814 feet per second (fps) and not less than 776 fps when tested as specified in 4.4.3.1 (see 6.9). The standard deviation of the lot shall not exceed 7.5 fps.

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship.

3.6.1 Propelling charge. The propelling charge shall be free of dirt, grease, or other foreign material.

3.6.2 Packaging. The packaging shall be free from foreign material. The boxes shall be assembled without undue binding and no flap shall project beyond the edge of the box when closed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following item in sample quantities as indicated.

Part Description	Drawing	Quantity
Charge, Propelling, M204	9312698	110

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4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078(AR).

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TABLE I - First article inspection

MIL-C-48867B (AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Charge, Propelling, M204				9312698
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination for defects	110		3.2	4.4.2.1
	Weight of propellant	10		3.2	4.5.2
	Gross weight of propelling charge	110		3.3	4.5.3
	Velocity (Proving Ground)	200*		3.4	4.4.3.1/4.5.4

NOTES:

* 200 is equivalent to the max. charge of 50 cartridges (4 charges per cartridge).

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of propelling charges shall contain:

a. Container assemblies of one lot interfix from one manufacturer.

b. Propellant from not more than one lot from one manufacturer.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

MIL-C-48867B (AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Charge, Propelling, M204				9312698
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Weight of propellant, in excess of three grains below min.	(a)(b)	100%	3.2	4.5.1
2	Gross weight of propelling charge, min.		100%	3.3	4.5.3
3	Container or container seal damaged to the extent that propellant shall escape		100%	3.2	Visual (4.5.5)
Major					
101	Weight of propellant from min. to and including three grains below min.	(a)(b)	100%	3.2	4.5.1
102	Weight of propellant, max.	(a)(b)	100%	3.2	4.5.1
103	Gross weight of propelling charge, max.		100%	3.3	4.5.3
104	Velocity (Proving Ground)	200/100(c)	1.0%	3.4	4.4.3.1/4.5.4
105	Container or container seal damaged but not to the extent that propellant shall escape		100%	3.2	Visual (4.5.5)
Minor					
201	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6.1	Visual

NOTES: (a) This inspection may be performed prior to loading. See 4.5.1 for selection of test method. (b) In the automated mode, this weight is the total loaded container weight minus the empty container weight for each loaded increment. (c) 200/100 is equivalent to the max. charge of 50/25 cartridges (4 charges per cartridge).

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48867B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Packing and Marking for Box, Fiberboard, Outer for Charge, Propelling, M204 or Container Assembly		9313703
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Charges incorrectly packed		0.40%	5.1	Visual
102	Barrier bag damaged (punctured)		0.40%	5.1	Visual
103	Marking improper (barrier bag, outer box)		0.40%	5.3	Visual
104	Improper seal (barrier bag, inner and outer box)		0.40%	5.1	Visual
105	Orientation stripe on corner or outside box improperly located (with respect to loading holes of charge)		0.40%	5.1	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48867B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Packing for Pallet	1 OF 1	9313727
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Orientation of boxes, improper		0.40%	5.2	Visual
102	Strapping loose		0.40%	5.2	Visual
103	Boxes damaged		0.40%	5.2	Visual
Minor	None defined				

NOTES:

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4.4.3 Testing.

4.4.3.1 Velocity.

4.4.3.1.1 First article. The sampling plan for cartridges required and procedure for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter M and an AQL of 1.0 percent. Also, if the standard deviation of the sample exceeds the lot requirement, the first article of propelling charges shall be rejected. The test shall be performed as specified in 4.5.4.

4.4.3.1.2 Production.

4.4.3.1.2.1 Begining with the first lot and continuing until three consecutive lots have complied with the applicable requirements specified. The sampling plan for cartridges required and procedure/acceptance criteria for this test shall be in accordance with MIL-STD-414, Section B, Table B-3 Code Letter M and an AQL of 1.0 percent. Also, if the standard deviation of the sample exceeds the lot requirement, the lot of propelling charges shall be rejected. The test shall be performed as specified in 4.5.4.

4.4.3.1.2.2 After three consecutive lots have met the criteria of 4.4.3.1.2.1 the sampling plan for cartridges for this test shall be in accordance with MIL-STD-414, Section B, Table B-3 Code Letter I, and an AQL of 1.0 percent. Also, if the standard deviation of the sample exceeds the lot requirement, the lot of propelling charges shall be rejected. The test shall be performed as specified in 4.5.4. In the event of a lot rejection the sampling plan of 4.4.3.1.2.1 shall be required beginning with the next lot and continuing until three consecutive lots of propelling charges have complied with the applicable requirements.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Weight of propellant.

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4.5.1.1 Manual method.

4.5.1.1.1 Method 1. The propellant shall be weighed. Weight limits should be established within the applicable weight requirements, depending upon the accuracy and precision of the weighing system to assure compliance with the charge weight requirements. Any charge of bulk propellant failing to meet the applicable requirement shall be classified defective and removed from the lot.

4.5.1.1.2 Method 2. Samples for this test shall be selected from propelling charges that have met the requirements of 3.3. A sample of 125 propelling charges shall be randomly selected during production for each shift. The propellant shall be removed from the charge, and weighed. If one or more critical defects or two or more major defects are found, the production represented by the sample shall be rejected.

4.5.1.2 Automated method. The contractor may elect to perform this operation with automated equipment and mandatory process controls. The automated equipment shall weigh the container and place this weight in memory and then weigh/check weigh the propellant before loading. After loading, the total weight of the container/propellant combination shall be weighed to verify complete loading by subtracting the increment weight (from memory) from the total weight to obtain the loaded propellant weight. In such cases, process control plans in accordance with Data Item DI-P-1604 must be submitted to the Technical Agency (see 6.6) for review of the automated operation.

4.5.2 Weight of propellant in charge, propelling (first article).

4.5.2.1 Destructive test. The propellant shall be removed from the charge, propelling and weighed. If any sample fails to meet the requirement it shall be classified defective and the first article sample shall be rejected.

4.5.2.2 Non-destructive (alternate procedure). The charge, propelling shall be weighed and the net weight of propellant determined by subtracting parts tare weight from the gross weight. If any sample fails to meet the requirement it shall be classified defective and the first article sample shall be rejected.

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4.5.3 Gross weight of charge, propelling. The loaded charge, propelling shall be weighed individually. If any assembly fails to meet the requirement, it shall be classified defective and removed from the lot.

4.5.4 Velocity test. This test shall be conducted at a government proving ground. The rounds shall be inert loaded to approximately 3.75 pounds including the weight of the dummy fuze. The rounds shall be assembled at maximum charge (i.e. with four charges). The rounds shall be conditioned at 70° ± 2.5° Fahrenheit and fired from a hard mount at an elevation of 800 mils, from an M224-60MM mortar tube for which the round is standard. Reference rounds shall be fired using reference standard components and a calibration lot of charge, propellant and ignition cartridges for velocity correction. The individual velocities and pressures shall be obtained and recorded. Velocity standard deviation and mean velocity shall be calculated to determine compliance with applicable requirements. A formal firing report shall be submitted (see 6.7).

4.5.5 Damaged propelling charge. To ascertain that a damaged propelling charge is to be classed as critical or major, the following procedure shall be used:

a. Place the damaged charge on a M27 fin, oriented in such a mammer that the propellant would be resting against the damaged surface.

b. The fin-charge assembly shall be hand-shaken vigorously three times.

c. Examination shall be made for propellant spillage - if any propellant is observed, the damaged propelling charge shall be classed as a critical defect; if no propellant is observed the propelling charge shall be classed as a major defect.

5. PACKAGING

5.1 Packaging. Packaging shall be as specified on dwg. 9313703.

5.2 Packing. Packing shall be as specified on dwg. 9313727.

5.3 Marking. Marking shall be as specified on dwg. 9313703 and dwg. 9313727.

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6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the 60MM Lightweight Company Mortar System (LWCMS).

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Command, US Army Armament Research and Development Command, ATTN: DRDAR-QAT-I, Dover, NJ 07801.

6.4 Data requirements. Deliverable data required by this specification is cited in the following paragraph:

Paragraph	Data Requirement	Applicable DID
4.5.1.2	Manufacturing Methods Reports (Modified)	DI-P-1604

(Copies of data item descriptions required by the contractor in connection with the specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.6 Submission of process control plans for approval. Submit plans as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAT-B, Dover, NJ 07801.

6.7 Submission of test data. Formal firing reports containing velocity test data and lot acceptability shall be sent to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAT-B, Dover, NJ 07801.

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6.8 Proving ground test summary.

Test	Sample Size	Requirements
First Article		
Velocity	200-Charge, Propelling, 50-60MM, M720 Inert Cartridges and Fuzes 17-60MM, M720 Reference Calibration Cartridges	See para. 3.4
Production Lots		
Velocity	200-Charge, Propelling 50-60MM, M720 Inert Cartridges and Fuzes 17-60MM, M720 Reference Calibration Cartridges	See para. 3.4
Velocity (after three successful lots)	100-Charge, Propelling 25-60MM M720 Inert Cartridges and Fuzes 9-60MM, M720 Reference Calibration Cartridges	

6.9 Velocity requirement. The velocity requirement in this specification is established for testing mortar cartridges fired from a hard mount (armor plate). Since all velocity assessments will be conducted from hard mounts, a correction factor applied to the assessed velocities will reflect the appropriate firing table velocities when a standard mortar base plate is used.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1310-A284)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-48867B	CHARGE, PROPELLING, M204 LOADING, ASSEMBLING AND PACKING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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